

Linking relationship quality and resourcefulness to group performance

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Abstract

Purpose – The purpose of this paper is to know that how group resources (internal and external) and the relationship quality among group members relate to group performance.

Design/methodology/approach – Given the normative nature of group performance, the study is carried out in a contrived environment. Participants were 204 master of business administration students who were allocated to 51 study groups. Data were collected in three waves and from two different sources, i.e., students and instructors. Data analysis was carried out by employing regression analysis and the bootstrapping procedure, i.e., PROCESS.

Findings – The results of this paper reveal that an individual-level internal resource, i.e. time, positively predicts group performance, while group-level internal resources, i.e., group composition and group members' experience, negatively predict group performance. Both external resources (external communication and instructor's support) are found to have a positive effect on group performance. The relationship quality among group members partially relates to group performance. Instructor's support as an external resource is found to moderate the relationship between only two aspects of relationship quality and group performance.

Practical implications – This study provides guidance to group members as to how they can utilize internal and external group resources and their relationship quality for enhancing their group performance. Managers in varied organizations can also utilize the findings of this study.

Originality/value – This study is unique in that it offers a new insight into internal and external resources and relationship quality, that is, from the perspective of group performance. The group resources included in the study are rarely found in the existing literature.

Keywords Relationship quality, External resources, Group performance, Internal resources

Paper type Research paper

Introduction

Mirroring each other, both business schools and business organizations pursue innovative efforts that could lead them to business excellence (Dahlgaard, 2014). A roller-coaster relationship exists between business schools and business organizations. There are some practices which the former learns from the latter and vice versa. However, both undertake some practices simultaneously, and group work is one of them. Observing how business organizations impart training to their managers and employees, business schools try to develop the future managers (students) for using group resources efficiently, maintaining good relationship quality among group members and thus, attaining high group performance.

According to McGrath (1991), different groups follow different paths and go through different stages to achieve high group performance. For instance, in order to foster students' understanding and make them capable of applying what they learn, group-based learning is



administered in business education. This is because in contemporary organizations business and management tasks are often performed in groups. Group-based learning is an innovative learning technique that improves group performance both at individual and group levels. In line with the above, the literature suggests that students' academic performance is enhanced, especially when they carry out study projects in groups (Zgheib *et al.*, 2010). Group-based learning facilitates information sharing and relationship quality among group members by establishing long-term relationship orientation. Overall, relationship quality helps group members achieve business excellence for their organization (Mashhadi *et al.*, 2014). Generally, group members complete the group project by utilizing the available internal and external resources.

While accentuating group intelligence, Woolley *et al.* (2015) called for exploring and examining the internal and external factors that could ensure high group performance. Notably, research on group performance thus far has rarely been conducted with respect to internal and external resources of groups. In view of the above, business education is considered essential in universities, especially in developing countries, as it is a source of producing developed human resources for local business organizations as well as for the international market (Fryer and Ogden, 2014). For this reason, universities have now shifted from passive learning to a group-based learning approach.

Drawing upon the resource-based view (Barney, 1991) together with resource dependence theory (Salancik and Pfeffer, 1977), the study includes both internal and external resources in the research model. Specifically, internal resources are studied at two levels, i.e., individual level ("time" and "finance" available to an individual group member) and group level (group formation – males, females or both – and group members' experience of working in group projects). External resources include supervisor's support and external communication (communication with members of other groups). In addition, four elements of relationship quality (information sharing, communication quality, long-term relationship orientation and satisfaction with the relationship) are included in the study.

Being mindful of the significance of efficient resource utilization and relationship quality for attaining high group performance, our study seeks to increase understanding of the factors that can influence group performance. More specifically, the study aims to answer the following questions:

RQ1. Do internal and external resources affect group performance?

RQ2. Does relationship quality among group members predict group performance?

RQ3. Do external resources interplay with the relationship between relationship quality and group performance?

Bearing in mind that business organizations reflect on business schools, especially with regard to group-based projects, for empirical testing of the study relationships we have focused on graduate students enrolled in the master of business administration (MBA) program of a business school. Since, our study is carried out in an artificial phenomenon, we discuss the relevant literature in the context of students' group projects.

Group resources and group performance

Being mindful of sustainability, contemporary educational institutions realize that they must be competitive. Toward this end, these institutions are restructuring their systems and policies. Likewise, business schools are preparing themselves for facing up to more challenges as globalization and change in business trends are on the rise. For instance, in order to prepare future managers who could truly fulfill the requirements of contemporary business organizations, business schools are employing advanced instructional designs. These instructional designs focus more on practicums, wherein students are put in

situations that resemble an actual business environment. Specifically, students are trained to become capable of working in groups. Therefore, the extant literature has witnessed a shift in the focus of business schools, that is, from individuals to teams (Yong *et al.*, 2014; Akbari *et al.*, 2016).

In relation to group performance, previous studies focused on a variety of areas, *inter alia*, intra-group conflict (Jehn and Mannix, 2001), workgroup diversity (Van Knippenberg *et al.*, 2004), social networks (Sparrowe *et al.*, 2001), group cohesion (Beal *et al.*, 2003), group beliefs and ability (Hecht *et al.*, 2002). Previous research works on group performance have also attempted to predict group outputs, and that while testing process-related assumptions (Davis, 1969; Steiner, 1972). However, the one that gained popularity among researchers is the research by Steiner (1972). His model focuses on achieving the optimum level of group performance by using idealized combinations of group members' resources.

There are different group resources, which according to resource-based view and resource dependence theory can be combined and orchestrated to produce the desired group performance. We categorize group resources into two sets, i.e., internal and external resources. For the accomplishment of group tasks, group members' reliance on one another cannot be ruled out. However, each group member's dependency on other group members can vary, that is, some group members need to work less dependently while others may have to work more collectively (Van der Vegt *et al.*, 2000, 2001). Therefore, as a part of their professional training, students' academic performance is evaluated based on the collective work accomplished by them (Zgheib *et al.*, 2010). Better said, the more the group members are resourceful at both individual and group levels, the better the group performance will be.

Internal resources (individual level)

This study considers that two internal resources at individual level, i.e., time and finance, can have the potential to positively predict group performance. Here, time refers to whether a group member is a full-time or a part-time student. According to Astin's (1984) involvement theory, there are less chances of success for students employed off-campus, as their work consumes their time and energy and also keeps them away from their academic assignments which are to be performed while being off-campus. The theory has gained empirical support that part-time students' number of working hours inversely relate to their degree of involvement in the assigned academic tasks (e.g. Furr and Elling, 2000). In contrast, students' number of hours spent on-campus, and that with teachers and peers, directly relates to their learning (Astin, 1984; Pescosolido, 2001).

Given the prevalent economic and social conditions in countries like Pakistan, where the present study has been carried out, the scarcity of employment opportunities in business environment means that employees remain in competition with each other. As a result of this, the youth continues education along with earning their livelihoods. Like other economies, muddying the waters of proper school-to-work transition, part-time jobs are witnessed to have rapid growth (Nothdurft, 2010). Because of divided attention between school and job, part-time students' performance is more likely to be different from full-time students. Generally, it is considered that part-time students have an edge over the full-time students, i.e., being well conversant with what is happening in the world of work they can contribute more toward the group accomplishments (Pescosolido, 2001). On the contrary, students who are enrolled as part-time have less opportunity to meet with teachers and peers and hence their performance remains low (Lundberg and Schreiner, 2004). However, the fact is that the performance is not only about knowing but doing as well. Therefore, it is expected that students devoting full-time to their studies can contribute to their group performance more than part-time students. Thus, we hypothesize that:

H1a. Full-time students can contribute to group performance more than part-time students.

Finance is another important internal resource. In current times, higher education has become a privilege that everyone cannot afford, especially in developing countries. From higher education institutions' point of view, students are attracted toward education by providing them with financial support in terms of scholarships and bursaries. This is also because numerous students leave education due to the lack of funds (Lau, 2003). Therefore, financial support is considered important not only for students' retention in academia, but for improving their academic performance too. Abraham and Clark's (2006) findings suggest that students who receive financial assistance tend to remain in an institution and achieve higher grades than their peers who are given admission to university on a self-finance basis.

Due to high competition in the job market, employees' desire to achieve excellence in their respective line of work and attain a competitive advantage over other job seekers. To achieve these goals, individuals opt for continuing their education, i.e., getting education while simultaneously doing jobs. However, maintaining equilibrium between the demands of school and job is a challenge for such university students as they cannot expend the requisite time on their studies. Therefore, to continue the education, students need funding through earnings or savings, tuition reimbursements by their employers, parental support and scholarships and loans and grants for higher education. It is expected that the students who get financial support can give more time to academic activities that in turn may help them perform more efficiently in their academic life (Faulk *et al.*, 2012). Generally, in higher learning, and particularly for group-based learning, from "forming groups" to "performing in groups," individuals' financial resources play a vital role. Building on the above, we hypothesize that:

H1b. The students being awarded a scholarship can contribute to group performance more than self-financing students.

Internal resources (group level)

According to the extant literature, group composition can have a significant impact on group performance (Moreland *et al.*, 2013). Given, the between-group differences are under control, homogeneous groups attain better performance than heterogeneous groups (Watson *et al.*, 1993). Concerning surface-level differences, a group can consist of males, females (homogeneous) or both (heterogeneous). In the Pakistani environment specifically, where the present study has been carried out, people feel reluctant to work with the opposite gender. A large population in the region maintains that females and males are supposed to work only inside and outside their homes, respectively. However, considering the growing awareness, changing social circumstances, expanding dimensions of economics, e.g., womenomics, females are now also being increasingly encouraged to work with men. Yet, there is still reluctance in sharing ideas and point of views among male and female colleagues, especially when they have to go out for some fieldwork. Likewise, female students in their study projects face similar problems. Therefore, it is assumed that in class projects, the more male group members there are, the higher the group performance will be, and vice versa (Ensley and Hmieleski, 2005). Hence, it is hypothesized that:

H2a. The greater the number of male students in a group is, the higher the group performance will be.

The second component of internal resources at group level is the experience of working in groups. It means that individuals have a willingness to explore, consider and tolerate both new and familiar ideas and experiences (Schilpzand *et al.*, 2011). Per general perception, group members' experience is assumed to positively predict group performance. However, there is also a contrasting viewpoint in the literature (e.g. Homan *et al.*, 2008) which asserts that group members' experience can negatively predict group performance. This is likely to happen when more experienced group members dominate other group members by using their

practical knowledge. This may further cause subdual of other members' ideas, creativity and innovativeness, which is against the spirit of working in groups, i.e., free sharing of ideas among group members and using collective wisdom. Thus, it is expected that:

H2b. The more the students' experience of working in groups is, the lower the group performance will be.

External resources

There are some external factors that can have a direct impact on group performance. We consider the following two factors more compelling: supervisor's support and external communication. Besides group leader and group members, the supervisor's support is instrumental to group-based learning which in turn can improve group performance (Metheny *et al.*, 2008), although the supervisor is not a member of the group (Black and Deci, 2000). In a group, students can have different backgrounds and different characteristics. This diversity can be well managed if the instructor support and intervention is at hand (Duffy and Shaw, 2000). The literature suggests that instructor's support helps students to increase their interest in academic activities, competence and self-regulation and it helps students to diffuse their anxiety while taking up tough tasks in the semester (Black and Deci, 2000). Therefore, it is hypothesized that:

H3a. Instructor's support positively predicts students' group performance.

Being useful in both work and study contexts, external communication, i.e., sharing of ideas with individuals outside the group, can contribute to group performance (Chermak and Weiss, 2005). In group-based learning, external communication can help group members mitigate the effects of misinformation, obtain the quality information and data they require for completing their study project and seek informal feedback from out-group fellows (Lovall, 2001). Having the requisite information fetched from outside the group, group members feel more inclined to share that information within the group that helps them reiterate the group's objectives and then compare with the group's achievements, further leading to overall group performance (Smidts *et al.*, 2001). Based on the above, it is hypothesized that:

H3b. External communication positively predicts students' group performance.

Relationship quality and group performance

The literature suggests a positive relationship between relationship quality and group performance (see e.g. Curkovic *et al.*, 2000; Rickards and Moger, 2000). However, the development of effective in-group relations depends on the psychological traits of team members (Randeree and Ninan, 2011). These traits help them share their knowledge, skills and cognitive ideas to enhance group performance (Dong *et al.*, 2008). Effective flow of information, for learning purposes, can be attained through high-quality relationships among group members (Akgun *et al.*, 2006).

Different determinants of relationship quality have been discussed in the literature. Four determinants of relationship quality, namely, cultural understanding, cultural interaction, communication competence and communication interaction have been identified by Griffith and Harvey (2001). Pressey and Selassie (2003) suggested that some managers perceive that factors such as language barriers in communication, geographic distance, political barriers, economic factors, time differences, industry barriers, technology barriers, infrastructure barriers and legal differences can influence intra-group relationships. Ural (2009) also identified four elements of relationship quality: information sharing, communication quality, long-term relationship orientation and satisfaction with the relationship. In the context of

the present study, the dimensions used by Ural (2009) are considered more relevant as students have to work in groups and communicate, and seek intergroup and intragroup support in order to complete the assigned projects.

First, information sharing requires that information should be shared openly within groups to achieve high group performance (Lages *et al.*, 2004). Information sharing is important for improving the relationship quality (Ural, 2009), performance (Lages *et al.*, 2004) and satisfaction of group members by the effective utilization of available information (Cannon and Homburg, 2001). It is expected that while working in groups, group members with more information can use their knowledge for better group performance. Second, communication quality refers to the process that maintains a relationship between different individuals or groups. While working in groups, communication flow depends on the amount of information shared within the groups. Notably, two-way information exchange is necessary for the efficient use of information (Ural, 2009). Both types of communication, formal and informal, exist within groups. It is an intrinsic element of relationship quality (Roberts *et al.*, 2003) that results in high group performance.

Third, long-term relationship orientation leads to sustainable competitive advantage for groups (Keyton, 2000). It involves goal sharing, cooperation and risk sharing while working in groups. Each group member expects that his/her performance is contingent upon relationship quality (Ural, 2009). In this way, instead of one group, more groups can benefit from each other leading to improved group performance. The final, the fulfillment of desired goals within the groups lead to satisfaction with the relationship (Anderson and Narus, 1990). It is a positive emotional state resulting from the assessment of relationship within individuals or groups (Katz *et al.*, 2004). More satisfied individuals have a better relationship quality (Ural, 2009). Based on the above, we hypothesize that:

H4. Relationship quality may positively predict group performance.

Moderating effect of external resources on relationship quality–group performance

External resources can interplay with the relationship quality, amplifying its effect on group performance. That is, support from instructors to complete assigned tasks, when combined with relationship quality, produces more effective group performance (Black and Deci, 2000). Instructor's support has a strong relationship with both relationship quality (Mullen and Tallent-Runnels, 2006) and external communication. When information is gathered and shared outside a group, it has a multiplying effect on relationship quality, thereby increasing group performance (Chermak and Weiss, 2005). Therefore, external communication is also considered to strengthen the relationship between relationship quality and group performance. Hence, it is hypothesized that:

H5a. Instructor's support moderates the association between relationship quality and students' group performance by strengthening the relationship.

H5b. External communication moderates the relationship between relationship quality and students' group performance by strengthening the relationship (Figure 1).

Method

Sample and procedures

The sample consisted of 204 students enrolled in an MBA program at a large public sector university in Islamabad, Pakistan. The study utilized 51 groups developed by the concerned course instructors on a random basis for group projects. The project assigned to the students was to be completed in a semester. Data were collected in three waves from two sources.

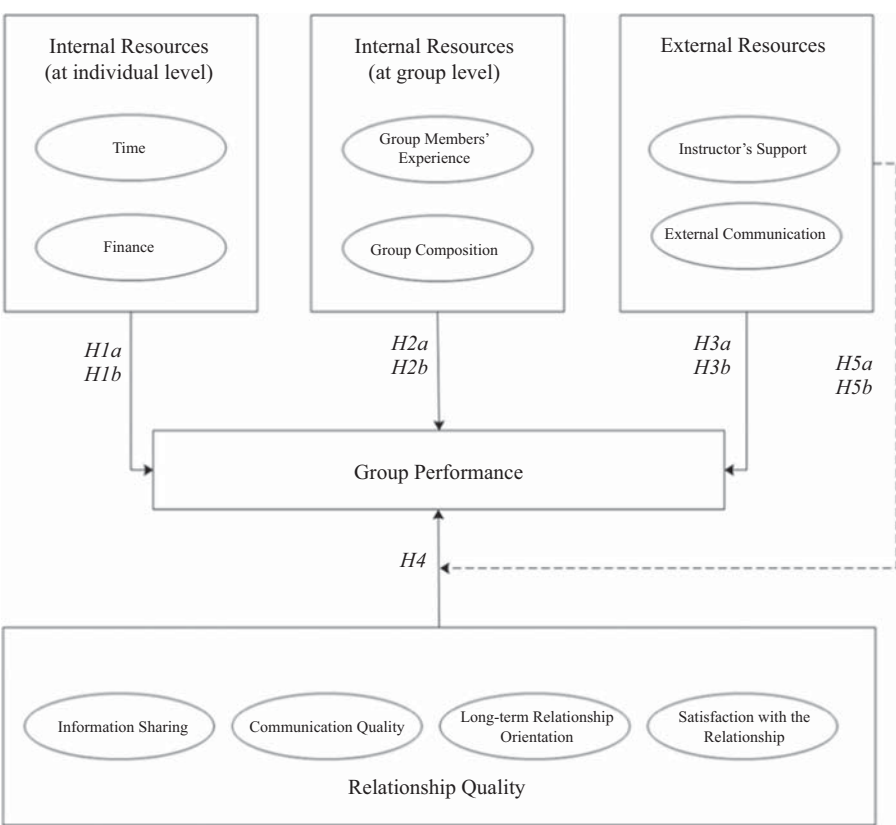


Figure 1.
The research model

That is, first at the start of semester (start of project), and second, after the completion of semester (end of project), the students filled two different questionnaires. In the third wave, actual project marks awarded by the instructors were collected. The group formation was based on the instructor's choice and without the consent of students. Two instructors were contacted for their help and support. After getting their consent, a brief about the study was presented to them and their input was also incorporated into the project assignment.

Questionnaires were first presented to and discussed with the course instructors for likely modifications, and information about the process of group formation was collected from them. A suitable time was discussed with them for the delivery of questionnaires in hard form to the respondents in the class. Respondents were assured about anonymity and confidentiality so that they could express their experiences without any hesitation. Ample time was given to students for going through the questionnaires. In order to obtain quality data, respondents' queries were addressed promptly. After 15 days, the first author visited the class and collected the available responses. The remaining questionnaires were collected from the instructors.

In the first wave, data about internal resources (at group level) and demographics were gathered. It included time- and finance-related data and personal information (time 1, questionnaire 1). This was the information that could be gathered initially as students had clarity about this while enrolling in a semester. In the second wave, data about constructs of external resources (instructors' support and external communication) and internal resources

(at group level), and relationship quality were gathered (time 2, questionnaire 2). In the third and the final wave, data about group performance were collected from the concerned course instructors. That is, students' actual performance was measured using their obtained marks in the project, which the concerned instructors awarded to them based on a predetermined criterion. There were 51 different groups of four respondents each, therefore, instructors awarded marks to individual students, which were summated for group performance.

Measures

While adapting measures, we considered two aspects, i.e., ensuring the consistency between measures included in the survey questionnaire and the respective construct of interest and incorporating slight modifications to help participants familiarize with the study context.

Relationship quality. Relationship quality was measured using 14 items from Ural (2009). This construct comprises four elements, therefore, amount of information sharing had three items (the sample item is: "Different approaches to complete the given project successfully were frequently discussed within the group"), communication quality (four items, "Group members continuously interacted to complete the project"), long-term relationship orientation (four items, "I focused on long-term goals in this relationship") and satisfaction with the relationship (three items, "My association with this group was successful one"). All items were measured using a five-point Likert type scale, ranging from 1 = strongly disagree to 5 = strongly agree.

Group performance. For measuring group performance, marks given by the instructors on the basis of group members' performance in the projects were utilized. These marks were used as they were actual and more authenticated than perceived group performance.

Internal resources. Internal resources at individual level include time and finance. Respondents were asked whether they are 1 = full-time students or 2 = part-time students. Regarding financial support, respondents were asked whether they are 1 = self-financed or 2 = scholarship recipients. Internal resources at group level include group members' experience and group composition. For measuring the former, respondents were asked that how many times they remained involved in group projects: 1 = less than three times, 2 = four times, 3 = five times, and 4 = six times or more. Whilst for measuring the latter, respondents were categorized into three groups: 1 = only males, 2 = only females and 3 = both.

External resources. External resources include instructors' support and external communication. For measuring the both these variables, respondents were asked to mention the level of instructors' help and external communication on the following scale: 1 = low, 2 = medium and 3 = high.

Results

Table I presents means, standard deviations and correlations.

Main effects

H1a–b pertain to the association between internal resources at individual level and group performance. Table II shows that the effect of time on group performance is positive significant ($\beta = 0.30$, $t = 3.11$, $p < 0.01$), whereas the effect of finance on group performance is not significant ($\beta = -0.10$, $t = -1.07$, ns). Thus, *H1a* is confirmed but *H1b* is not supported.

H2a–b pertain to the association between internal resources at group level and group performance. As expected, group composition ($\beta = -0.16$, $t = -2.30$, $p < 0.05$) and group member's experience ($\beta = -0.15$, $t = -2.25$, $p < 0.05$) negatively predict group performance. Thus, *H2a* and *H2b* are accepted.

Table I.
Means, standard
deviations,
intercorrelations and
reliability coefficients

Variables	M	SD	1	2	3	4	5	6	7	8	9	10
(1) Time	1.52	0.45	–									
(2) Finance	1.56	0.52	0.71***	–								
(3) Group composition	2.53	0.67	–0.12***	–0.04	–							
(4) Group members' experience	2.95	0.95	–0.05	0.09	0.09	–						
(5) External communication	1.50	0.50	0.29***	0.19**	–0.34***	0.08	–					
(6) Instructor's support	1.50	0.50	0.39***	0.17*	0.02	–0.31***	0.44***	–				
(7) Information sharing	3.73	0.42	–0.05	0.01	0.08	0.26***	–0.21**	–0.29***	0.64			
(8) Communication quality	3.78	0.38	0.01	–0.09	–0.55***	0.26***	0.01	–0.03	0.11	0.64		
(9) Long-term relationship orientation	3.53	0.41	0.09	–0.03	–0.17*	–0.20**	0.32***	0.17*	0.23***	0.23***	0.74	
(10) Satisfaction with the relationship	3.55	0.41	–0.16*	–0.20**	0.12***	–0.46***	–0.19**	0.02	–0.13***	0.35***	0.31***	0.72
(11) Group performance	3.05	1.25	0.23***	0.11	–0.17*	–0.17*	0.60***	0.54***	–0.34***	0.04	0.11	–0.10

Notes: $n = 204$. Cronbach's α coefficients are reported in *italics*. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.10$

Predictors	β	Group performance (marks)		Relationship quality
		R^2	$F(df_1, df_2)$	
Internal resources (individual)		0.06	6.21(2, 201)**	635
Time	0.30**			
Finance	-0.10 ^{ns}			
Internal resources (group)		0.05	5.66(2, 201)**	
Group composition	-0.16*			
Group members' experience	-0.15*			
External resources		0.46	84.35(2, 201)***	
External communication	0.45***			
Instructor's support	0.35***			
Relationship quality		0.22	14.12(4, 199)***	
Information sharing	-0.45***			Table II. Regression results
Communication quality	0.13*			
Long-term relationship orientation	0.28***			
Satisfaction with the relationship	-0.29***			

Notes: $n = 204$. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

H3a–b pertain to the association between external resources and group performance. As expected, the results reveal that instructor's support ($\beta = 0.35$, $t = 7.69$, $p < 0.001$) and external communication ($\beta = 0.45$, $t = 6.00$, $p < 0.001$) have a significant positive effect on group performance. Hence, *H3a* and *H3b* are supported.

H4 states that relationship quality positively predicts group performance. The results reveal that all the elements of relationship quality, i.e., information sharing ($\beta = -0.45$, $t = -6.85$, $p < 0.001$), communication quality ($\beta = 0.13$, $t = 1.91$, $p < 0.10$), long-term relationship orientation ($\beta = 0.28$, $t = 4.04$, $p < 0.001$) and satisfaction with the relationship ($\beta = -0.29$, $t = -4.10$, $p < 0.001$) appear to have significant relationship with group performance. However, contrary to our expectations, two elements (i.e. information sharing and satisfaction with the relationship) had negative relationship with group performance. Thus, *H4* is partially supported.

Moderating effects

H5a–b pertain to the moderating effects of external resources, i.e., instructor's support and external communication, on the relationship between relationship quality and group performance. In order to run moderation analysis, we employed Preacher and Hayes' (2008) bootstrapping procedure (5,000 iterations, bias-corrected, 95% confidence intervals (CI)). We ran eight moderated models. In models 1–4, we analyzed the moderating effect of instructor's support on each factor of relationship quality, and in models 5–8, we analyzed the moderating effects of external communication on each factor of relationship quality. Of eight moderated models only the following two models provided evidence of moderating effects.

First, the direct paths between information sharing ($B = 1.17$, $p < 0.001$) and instructors' support ($B = -0.74$, $p < 0.001$), and group performance were highly significant, and the effect of the interaction term (information sharing $\times$ instructor's support: $B = -0.86$, $p < 0.05$; $\Delta R^2 = 0.02$, $F(1, 200) = 5.08$, $p < 0.05$) on group performance was also significant. In addition, the interaction effect at +1SD on mean was $CI_{95\%}(-1.79, -0.54)$ and that at -1SD on mean was $CI_{95\%}(-0.73, 0.10)$. For conditional effect, see Panel A of Figure 2.

Second, the direct path between long-term relationship orientation ($B = 1.32$, $p < 0.001$) and group performance was significant, and instructors' support ($B = 0.20$, ns) and group performance was insignificant; however, the effect of the interaction term (long-term

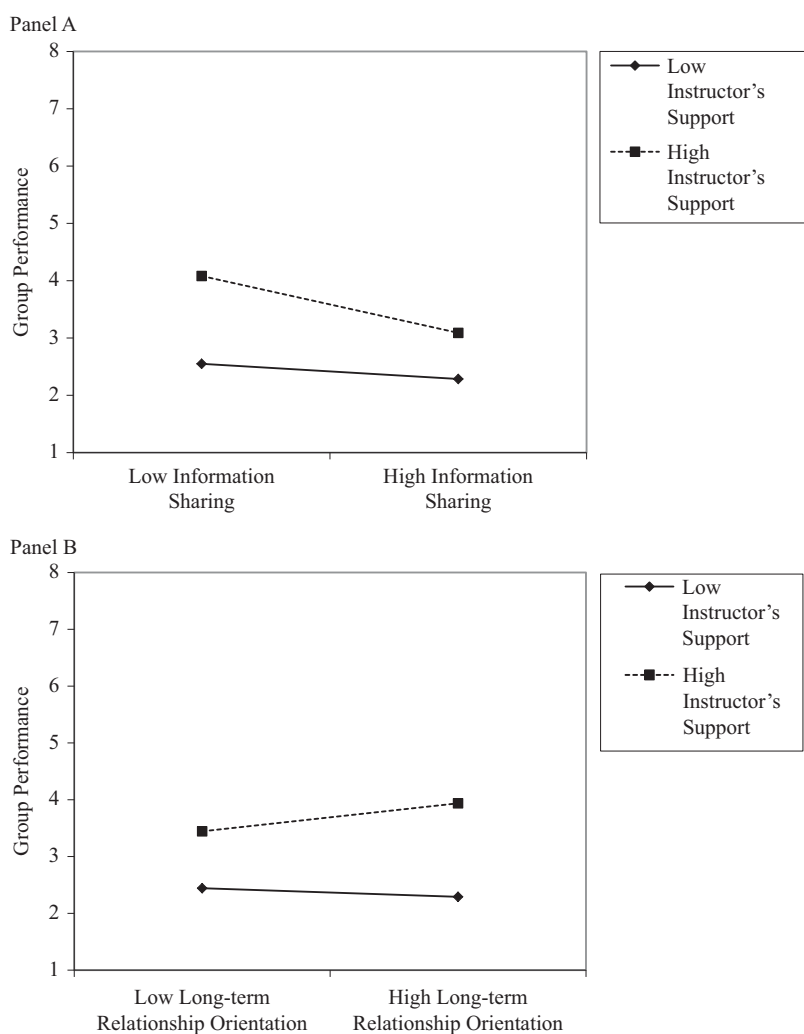


Figure 2.
Moderating effects

relationship orientation $\times$ instructor's support: $B = 0.78, p < 0.05; R^2 = 0.31, F(3; 204) = 30, p < 0.001$) on group performance was significant. In addition, the interaction effect at $+1SD$ on mean was $CI_{95\%}(0.03, 1.23)$ and that at $-1SD$ on mean was $CI_{95\%}(-0.61, 0.25)$. For conditional effect, see Panel B of Figure 2.

Moderating effects in rest of the six models were insignificant. For details, see Table III.

Discussion

There are certain factors that contribute to group performance. Theory of resource-based view suggests the use of group resources (internal) in such a combination that these could create a synergy effect in order to increase workgroup performance. However, according to resource dependence theory, while utilizing group resources group leaders and members should bear in mind that certain external resources of the group can influence group internal environment both positively and negatively.

Predictors	DV: group performance (marks)					
	<i>B</i>	SE	ΔR^2	$F(df_1, df_2)$	Conditional Effects CI _{95%} +1SD on <i>M</i>	-1SD on <i>M</i>
Model 1			0.02	5.08(1, 200)*	[-1.79, -0.54]	[-0.73, 0.10]
Information sharing	1.17***	0.15				
Instructor's support	-0.74***	0.19				
Information sharing × Instructor's support	-0.86*	0.38				
Model 2			0.00	0.71(1, 200) ^{ns}	[-0.17, 0.87]	[-0.56, 0.59]
Communication quality	1.36***	0.15				
Instructor's support	0.18 ^{ns}	0.20				
Communication quality × Instructor's support	0.33 ^{ns}	0.39				
Model 3			0.01	4.10(1, 200)*	[0.03, 1.23]	[-0.61, 0.25]
Long-term relationship orientation	1.32***	0.15				
Instructor's support	0.20 ^{ns}	0.19				
Long-term relationship orientation × Instructor's support	0.78*	0.39				
Model 4			0.00	0.01(1, 200) ^{ns}	[-0.90, 0.15]	[-0.81, 0.15]
Satisfaction with the relationship	1.36***	0.15				
Instructor's support	-0.35 ^{ns}	0.18				
Satisfaction with the relationship × Instructor's support	-0.04 ^{ns}	0.36				
Model 5			0.00	0.37(1, 200) ^{ns}	[-1.00, -0.13]	[-1.25, -0.29]
Information sharing	1.37***	0.14				
External communication	-0.67***	0.16				
Information sharing × External communication	0.20 ^{ns}	0.33				
Model 6			0.01	2.43(1, 200) ^{ns}	[-0.33, 0.43]	[-0.24, 2.87]
Communication quality	1.48***	0.14				
External communication	0.68 ^{ns}	0.41				
Communication quality × External communication	-1.26 ^{ns}	0.81				
Model 7			0.00	0.72(1, 200) ^{ns}	[-0.59, 0.31]	[-1.02, 0.11]
Long-term relationship orientation	1.57***	0.15				
External communication	-0.30 ^{ns}	0.18				
Long-term relationship orientation × External communication	0.31 ^{ns}	0.37				
Model 8			0.00	1.62(1, 200) ^{ns}	[-0.47, 0.31]	[-0.29, 1.22]
Satisfaction with the relationship	1.5***	0.14				
External communication	0.19 ^{ns}	0.21				
Satisfaction with the relationship × External communication	-0.55 ^{ns}	0.43				

Notes: *n* = 204. **p* < 0.05; ****p* < 0.001

Table III.
Moderation results

In the present study, both internal and external resources are included. Internal resources are further divided into resources at individual and group levels. Individual resources that are inside the group are time and finance. Time is related to whether a student is enrolled as a full-time or part-time student. According to results of our study, time is found to have a significant positive effect on group performance, supporting previous studies (e.g. Lucas and Lammont, 1988). Full-time students have no other obligations, but to devote full attention to their studies that contributes to such students' individual performance as well as performance of the group they work for. However, our data could not confirm the contribution of finance toward group performance. Previous studies suggest that students

remain focused on their studies as long as the costs of their study are met no matter how these are met (see e.g. Faulk *et al.*, 2012).

We studied the role of two group-level internal resources toward group performance, i.e., group composition and group members' experience. The results of our study confirm that both these resources negatively predict group performance. According to Zhan *et al.* (2015), gender-based group composition significantly influences group performance. They have observed that males perform better in mix-gender groups and females can perform equally whether group comprises all female members or mixed gender. Moreover, males prefer to join groups having a mix-gender, while females prefer single-gender groups. Gender differences occur on attitude rather than learning achievement. For group formation and improved performance, it is required that group members demonstrate mutual interest and understanding with each other.

Woolley *et al.* (2015) studied collective intelligence and found it an ability of groups to perform multiple tasks. According to them, group composition and group interaction can potentially influence group intelligence. In contrast, some other researchers suggest that group composition may lead to loss of trust and compatibility leading to decreased group performance (see e.g. Ensley and Hmieleski, 2005). According to them, group composition is important for understanding others' viewpoints in the group. Groups having individuals with more experience tend to have decreased performance as they attempt to impose their ideas on others, and thus, either they make decisions on their own or influence others for the purpose. Moreover, highly experienced group members attempt to accomplish tasks without seeking others' input. This reduces the chances of others' learning, and may demotivate them, resulting in overall low group performance (Homan *et al.*, 2008).

External resources include external communication and instructor's support. The results of our study confirm that both have a positive significant relationship with group performance. External communication is the communication with other groups. The more the external communication is, the more the level of information sharing and knowledge transfer will be. As a result, group performance increases (Smidts *et al.*, 2001; Lovall, 2001; Chermak and Weiss, 2005). As regards instructors' support, if instructors are supportive and guide students, the group of students can increase their group performance (Black and Deci, 2000; Duffy and Shaw, 2000).

We examined the relationship between four dimensions of relationship quality and group performance. All four dimensions were found to have a significant impact on group performance, but contrary to our expectations, two of them, i.e., information sharing and satisfaction with the relationship, appeared to have an inverse relationship. The remaining two, i.e., communication quality and long-term relationship orientation, confirmed their direct relationship with group performance. This means that when the quality of communication increases, within or across groups, the understanding of information becomes clearer. Consequently, the clearer the ideas are, the higher is the group performance (Roberts *et al.*, 2003). Likewise, when group members intend to develop long-term relationships amongst them they try to work collectively, share information and make communications successful. As a result, the group performance increases (Keyton, 2000; Ural, 2009).

While examining the moderating effect of external resources on the relationship between dimensions of relationship quality and group performance, we found that external communication did not interact at all. It could be because in our study we assigned every group a unique project. Maybe because of this, external communication could not moderate the relationship between relationship quality and group performance, as ideas brought into each group, from outside the groups, may not be relevant to the respective project (Katz *et al.*, 2004). However, the other external resource, instructor's support interacted with

the above relationship, but only in case of two dimensions of relationship quality, i.e., information sharing and long-term relationship orientation.

Based on the results of our study, instructor's support moderates the relationship between information sharing and long-term relationship orientation. While moderating information sharing and group performance, the instructor's support strengthens the relationship. When instructor's support is involved in this relationship, information sharing becomes more effective because of verification of information shared within or across groups (Cannon and Homburg, 2001; Lages *et al.*, 2004; Ural, 2009). However, if information sharing already has a negative effect on group performance then it can further increase by instructor's support. Therefore, instructor's support needs to be specific. That is, instructors should work toward resolving intra-group conflicts. Instructor's support also moderates the relationship between long-term relationship orientation and group performance by further strengthening the relationship. Instructors facilitate this relationship (Black and Deci, 2000) and hence students are more prone to achieve good group performance due to their inclination toward long-term relationship. As a result of long-term relationship orientation together with instructor's support and guidance, group members help each other in completing projects by assisting and guiding them throughout their projects and increasing their performance (Anderson and Narus, 1990; Ural, 2009).

Implications

This paper has implications for both academics and non-academics. That is, it contributes to the current literature by examining the different effects of internal and external organizational factors on group performance along with components of relationship quality. Group performance in higher education has been a slightly ignored area. Therefore, this study provides an additional insight about how to work in groups. In line with this, the findings of this study can help academics to consider group formation (based on gender and experience) for enhancing the learning experience of students. Moreover, this study will help instructors to improve their supportive role toward the completion of group projects.

Our findings can also be helpful for instructors who may consider the available resources both at individual and group levels while making groups. Furthermore, external resources can be considered while making groups, as these can help group members in order for working with like people to produce more output and increasing chances of group learning. This study can help practitioners, e.g., HR managers, line managers, understand the important role of group formation toward group performance. Moreover, it will also help them consider the internal organizational resources and external organizational resources for the achievement of group goals.

Limitations and future research

Alongside the above implications, we realize that the study has some limitations as well. The study is conducted in a relatively established university. When conducted in less developed ones, the results might be different. Besides studying the proposed relationships from business school's perspective, these can also be tested in some organizational setting. Parallel to supervisor's support, overall, administrative support can also be studied in the future. Moreover, instructors can also be asked about support provided by them to students.

Conclusions

This research has been conducted to study the role of group resources and relationship quality toward group performance. Resources at both levels (i.e. internal and external) have

been studied. The study is original in the sense that it has considered resources with respect to group performance and relationship quality for the first time under the tenants of resource dependence theory. Components of resources are already available in literature, but they have not been studied from the given perspective. The study presented interesting findings indicating that most of the variables under study lead to the enhancement of group performance. This insight can be helpful for instructors when they develop groups of students by taking into consideration the factors that have studied to enhance the group performance. Likewise, in organizations these findings can also be useful for managers, especially for developing teams.

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